Supplementary Material: Benchmarking LLM Integration Strategies for MCDA-Assisted Household Energy Decision Support

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S1. Extended Methods

This part of the supplementary material supports the main paper's methodology: full parameter-visibility tables, verbatim prompt texts, the retrieved-exemplar rendering, the RAG corpus description, a worked example with failure analysis, the Pennsylvania rate structure, and the utility-function derivations.

0.1 Full Parameter Comparison Tables

Table S1.1 lists the Appliance and Shower parameter visibility, following the same LLM-visible/calculator-visible split shown for HVAC in Table 1 (Section 3.2) of the main paper.

Table S1.1: Scenario parameters available to LLM architectures vs. ground-truth calculators (Appliance and Shower).

Appliance

Parameter	LLMs	Calc.
Appliance Type	T	✓
Baseline Runtime	T	✓
Appliance Age Tier	✓	—
kWh per Cycle	—	✓
TOU Period / Utility Rate	—	✓
Monthly Cycles	—	✓
Household Size	✓	✓
Housing Type	✓	✓
Location	✓	✓
Utility Budget	✓	✓

[T] T = free-text input; A = inferred from alternative.

Shower

Parameter	LLMs	Calc.
Location	✓	✓
Outdoor Temp	✓	✓
Inlet Water Temp	—	✓
Heater Setpoint	—	✓
Flow Rate Tier	✓	—
GPM	—	✓
Duration (min)	A	✓
Household Size	✓	✓
Tank Size	—	✓
Utility Budget	✓	✓
Housing Type	✓	✓

0.2 Full Prompt Texts

Full verbatim text of the system and extraction prompts summarized in Section 3.6 of the main paper, in the order the architectures appear there.

0.3 $\mathcal{A}_{\mathrm{D}}$ System Prompt

```
System prompt:

You are an expert household decision analyst specializing in Multi-Criteria Decision Analysis (MCDA).
You consistently utilize all information given in the scenario context. Score alternatives on four criteria using t

HVAC:
- energy_cost: good when the setpoint demands little from the system given outdoor conditions; moderate when the sy
- environmental: good when the system runs efficiently; moderate when runtime and load are average; poor when high
- comfort: good when the setpoint feels comfortable for the outdoor temperature; moderate when slightly outside typ
- practicality: good when the setpoint is easy for the user and system to maintain without strain; moderate when bo

Appliance (scheduling):
- energy_cost: good during off-peak rate hours; moderate in shoulder periods; poor when scheduled during peak prici
- environmental: good when grid emissions are low (typically overnight); moderate during shoulder hours; poor when
- comfort: good when the appliance is available with little or no delay; moderate when availability is pushed later
- practicality: good when scheduling is easy to remember and socially unobtrusive; moderate when timing requires sc

Shower (duration):
- energy_cost: good when short; worsens continuously as duration increases.
- environmental: good when water volume used is low; worsens continuously as duration increases.
- comfort: good near a typical shower length; poor when too short to feel adequate or long enough to feel wasteful.
- practicality: good when the duration is sustainable as a daily habit; poor when too short to realistically mainta

Return ONLY: {"energy_cost": X, "environmental": X, "comfort": X, "practicality": X} where each X is between 0.0 an
```

The user prompt provides the scenario context and asks the model to score one alternative at a time, including the other alternatives as reference so that it doesn't assign the same scores to all alternatives.

0.4 $\mathcal{A}_{\mathrm{E}}$ System Prompt

```
System prompt:

You are an expert household decision analyst specializing in Multi-Criteria Decision Analysis (MCDA).
You consistently utilize all information given in the scenario context. Score alternatives on four criteria using t
1. Energy Cost: Lower energy costs = higher score.
2. Environmental Impact: Lower emissions = higher score.
3. Comfort: Higher user comfort = higher score.
4. Practicality: Easier to implement/maintain = higher score.

Return ONLY: {"energy_cost": X, "environmental": X, "comfort": X, "practicality": X} where each X is between 0.0 an
```

The user prompt presents the target scenario, the alternative to score, and the retrieved exemplar—with household-reported parameters, engineering values, and the four ground-truth scores plus MAVT aggregate and rank—instructing the model to "consider how this specific alternative performs given the scenario context."

0.5 $\mathcal{A}_{\mathrm{H}}$ Extraction Prompt

```
Extraction prompt:

You are a household-systems engineer. You base all assumptions off of given information. Understand how each parame

SCENARIO:
{scenario_text}

QUESTION: {question}

INSTRUCTIONS:
1. Identify the Decision Type: HVAC, Appliance, or Shower.
2. Estimate ONLY the parameters listed for that type below.
3. Every listed parameter is mandatory. If a value is not stated, it is mandatory to reasonably estimate it from th
4. Return ONLY valid JSON in the exact structure shown.

For HVAC decisions:
{"decision_type": "HVAC",
  "calculator": "HVACGroundTruthCalculator",
  "parameters": {"r_value": <number>,
    "seer": <number>,
    "hvac_age": <number>,
    "occupancy_context": "occupied_all_day |
      unoccupied_<hours> | occupied_sleep"}}

For Appliance decisions:
{"decision_type": "Appliance",
  "calculator": "ApplianceGroundTruthCalculator",
  "parameters": {"appliance": "Dishwasher | Washer | Dryer",
    "kwh_per_cycle": <number>,
    "baseline_time": "<the time it currently is, e.g. 7pm, 8am>"}}

For Shower decisions:
{"decision_type": "Shower",
  "calculator": "ShowerGroundTruthCalculator",
  "parameters": {"gpm": <number>,
    "tank_size": <number>,
    "water_heater_temp": <number>}}

Return ONLY the JSON, no explanation.
```

0.6 Retrieved Exemplar Rendering

A retrieved HVAC exemplar (engineering parameters and ground-truth scores included) is rendered as:

```
RELEVANT SIMILAR SCENARIOS WITH EXPERT SCORES:

Example 1:
  Question: What temperature should I set my thermostat to?
  - Location: Chicago
  - Outdoor Temp: 95 deg F
  - Square Footage: 2000 sqft
  - Insulation: Medium
  - Household Size: 3 occupants
```

```
  - Housing Type: Single-Family
  - House Age: 6-10 years
  - R-Value: 13
  - SEER: 14
  - HVAC Age: 8 years
  - Utility Budget: $200/month
  Expert scores:
  * 76 F: Energy Cost: 0.4/1, Environmental: 0.3/1,
    Comfort: 0.9/1, Practicality: 0.8/1
    | MAVT: 0.58, Rank: 2
  * 78 F: Energy Cost: 0.6/1, ...
Use these examples as reference, but score based on
the specific scenario above.
```

0.7 RAG Corpus Description

The retrieval-augmented generation (RAG) corpus consists of 90 unique scenarios (35 HVAC, 35 Appliance, 20 Shower) that are fully disjoint from the 195 test scenarios. Each RAG scenario has three alternatives, yielding 270 total rows across the three source files (HVACRagScenarios.xlsx, ApplianceRAGScenarios.xlsx, ShowerRAGScenarios.xlsx).

Scenarios were generated from the same parameter distribution as the test set but kept separate to prevent any test scenario from appearing in the retrieval index. A combinatorial audit ensured that no parameter combination was underrepresented: each categorical cross-cell (insulation tier $\times$ housing type, appliance type $\times$ flow-rate band, etc.) contained at least one scenario, and the 90 RAG scenarios were drawn from this audited matrix.

The retrieval index was built using the `sentence-transformers/all-MiniLM-L6-v2` embedding model (Reimers and Gurevych, 2019), producing 384-dimensional dense vectors. At query time, the single most similar scenario ($k = 1$) is retrieved by cosine similarity and rendered as a full worked example in the LLM prompt. The embedding string includes all household-reported parameters (location, square footage, insulation tier, household size, housing type, utility budget, outdoor temperature, house age) plus band labels for age and flow rate, but excludes engineering parameters and ground-truth scores, which appear only in the display metadata of the retrieved exemplars.

The disjointness of test and RAG sets was verified by construction (the rebuild script exports them from separate master-sheet partitions) and is enforced at query time because test scenarios are used as query vectors against an index that stores only RAG scenarios.

0.8 AH Worked Example and Failure Analysis

0.9 Worked Example: $\mathcal{A}_{\mathrm{H}}$ HVAC Scenario

This subsection traces one HVAC scenario end-to-end through the $\mathcal{A}_{\mathrm{H}}$ pipeline. The full step-by-step arithmetic (thermal load, EER conversion, energy/cost/emissions, budget penalty, comfort and practicality scores, value-function transformation) is reproducible by running `Ground Truth Calculators/HVACGroundTruthCalculator.py` in the project repository on the scenario below; only the inputs and the final result are shown here.

The scenario: Philadelphia, PA; outdoor temperature 95 °F (cooling season); 2,200 sq ft single-family home, medium insulation, house age 15 years; household size 4; \$200/month utility budget; alternatives of 72 °F, 76 °F, and 80 °F. $\mathcal{A}_{\mathrm{H}}$'s extraction step returns $R = 15$, SEER $= 13$, hvac_age = 10, and `occupancy_context = occupied_all_day`, each validated against the physically admissible bounds of Section 3.6 of the main paper before being merged with the household-reported fields above and the fixed electricity rate of \$0.19/kWh to produce the full scenario dict passed to `HVACGroundTruthCalculator`.

Carrying that scenario through Eqs. (6)–(11) of the main paper for all three alternatives yields:

Setpoint	v^*_{cost}	v_{env}	v_{comfort}	v_{prac}	**MAVT** s_j
76 °F	0.11	0.30	1.00	0.85	**0.466**
80 °F	0.17	0.37	0.67	0.85	0.441
72 °F	0.06	0.24	0.67	0.78	0.352

The recommended setpoint is 76 °F, which coincides with the ASHRAE 55 comfort optimum—the scenario was designed so the ground-truth calculator's tent-function peak and the budget-constrained energy–comfort tradeoff agree. The 80 °F alternative ranks second despite lower energy cost, because the comfort penalty at four degrees above the optimum outweighs its energy savings under the given weight vector. The 72 °F alternative ranks last: it imposes the highest cost, exceeds the \$200 monthly budget (triggering the exponential tier of $P(u)$), and offers no comfort advantage over 76 °F.

0.10 GPT-OSS Extraction Failure Analysis

The $\mathcal{A}_{\text{H}}$ architecture validates each extracted JSON before passing it to the ground-truth calculator. A scenario is recorded as an extraction failure when: the LLM response is not valid JSON; a required parameter is absent; a numeric parameter parses as non-finite or falls outside its physically admissible range; or the extracted decision type does not match the scenario's known type.

GPT-OSS 20B exhibited a 33.4% extraction failure rate on $\mathcal{A}_{\text{H}}$ HVAC scenarios (12.0% across all decision types), triggered entirely by invalid parameters. All of its $\mathcal{A}_{\text{H}}$ failures were HVAC scenarios; Appliance and Shower extraction succeeded in every run. In a typical failure the model returns well-formed JSON but estimates parameters outside the validation bounds of Section 3.6 of the main paper—for example, a SEER rating of 5 against a lower bound of 6.0. The validation layer detects the out-of-range value, flags the scenario as an extraction failure, and assigns the sentinel value (1928) to all four criterion scores for that scenario.

0.11 Pennsylvania Time-of-Use Rate Structure

Table S1.2 lists the six Pennsylvania utilities used by the appliance ground-truth calculator, with their TOU peak/off-peak windows and energy rates (\$/kWh). All scenarios use weekday summer rates year-round; seasonal peak-window shifts (e.g., PPL winter 4–8pm vs. summer 2–6pm) and weekend off-peak pricing are not modeled (see Section 3.4 of the main paper for discussion).

PECO (Rate R-TOU) (**?**) and PPL (Rate RTS) (**?**) use a 4-hour afternoon peak window (2–6pm weekdays). The three FirstEnergy utilities (West Penn, Penelec, MetEd) use a 7-hour peak window (2–9pm weekdays), with rates derived from the Jun-2025 Price-to-Compare (PTC) multiplied by approved multipliers (FirstEnergy Pennsylvania Electric Company, 2025, 2026). Duquesne uses a flat default rate (Rate RS) (**?**); an optional TOU schedule (peak 3–9pm) exists but is not applied in the benchmark because the majority of Duquesne customers remain on the flat rate.

Table S1.2: Pennsylvania utility TOU rate structure. Peak window is 24-hour clock (start, end). All rates in \$/kWh. The residential electricity rate of \$0.19/kWh used for HVAC cost normalization is the statewide bundled baseline (Administration, 2025; Pennsylvania Public Utility Commission, 2025).

Utility	**Peak Window**	**Peak Rate**	**Off-Peak Rate**
PECO	2–6pm	0.3200	0.0760
PPL	2–6pm	0.1600	0.0700
West Penn	2–9pm	0.1720	0.0880
Penelec	2–9pm	0.1850	0.0930
MetEd	2–9pm	0.2030	0.1000
Duquesne	2–9pm	0.1375	0.1375

0.12 Electricity Rate Provenance

For HVAC, cost percentiles reflect active-conditioning alternatives. A setpoint near outdoor temperature produces zero decision-period load and collapses to \$0, which makes the lower bound go to zero. The cost is estimated at \$0.19/kWh, which represents a bundled retail rate combining generation, transmission, distribution, monthly service fees, riders, and state taxes. Form EIA-861 reports a 2024 Pennsylvania statewide average price of 17.77¢/kWh based on total utility revenue divided by sales (Administration, 2025). However, in urban and suburban Philadelphia, the Pennsylvania Public Utility Commission 2025 Rate Report lists an average monthly PECO residential bill of \$135.29 for 702 kWh, establishing an effective bundled price of 19.27¢/kWh (Pennsylvania Public Utility Commission, 2025). Approved PUC rate settlements increase representative 700 kWh PECO bills to \$149.43 in 2025 (21.3¢/kWh) and \$152.13 in 2026 (21.7¢/kWh) (Pennsylvania Public Utility Commission, 2024). The \$0.19/kWh baseline provides a conservative, empirical rate for Pennsylvania decision modeling. Excluding zero-load points, the realized 8-hour cost distribution at Pennsylvania's bundled residential rate (\$0.19/kWh) (Administration, 2025; Pennsylvania Public Utility Commission, 2025, 2024) spans \$0.38–\$3.29. An Off option receives top cost score via decreasing-value normalization. These endpoints match residential HVAC efficiency benchmarks (Economics, 2013; Parker, Sherwin, Stedman and Anello, 2018; **?**; **?**).

0.13 Comfort and Practicality Utility Function Derivations

This appendix contains the piecewise value function derivations for the Comfort and Practicality criteria across all three decision types. These utility functions are applied to the true engineering parameter values (R-value, GPM, appliance age in years) rather than the homeowner-facing band labels (insulation, flow rate, age band) that appear in the LLM prompts and scenario description sheets.

0.14 Comfort

0.14.1 Appliance Comfort

Comfort is a piecewise decay from 10 at zero delay, with appliance-specific tolerance ceilings of 12 hr (dishwashers), 8 hr (washers), and 6 hr (dryers) (Paetz, Dutschke and Fichtner, 2012; Waseem, Lin, Liu, Sajjad and Aziz, 2020), plus an additive noise penalty scaled by housing type for runs after 10 pm (U.S. Environmental Protection Agency, 1974) and a household-size penalty. Apartment and Condo units receive the largest noise multiplier ($\kappa = 1.5$, shared walls transmit vibration directly), Single-family homes the smallest ($\kappa = 0.8$, isolated structure), with Townhouse and Rowhouse intermediate ($\kappa = 1.2$). The household-size penalty $\sigma(n)$ steps from 0.0 for fewer than 3 occupants, to 0.8 for 3–4, to 1.5 for 5 or more, reflecting the empirical finding that multi-person households experience disproportionate annoyance from delayed appliance cycles (Waseem et al., 2020). Delays are computed as the minimum circular distance from the baseline time t_0 on a 24-hour clock (avoiding wrap-around artifacts), i.e.,

$$d(t, t_0) = \min\big((h(t) - h(t_0)) \bmod 24,\ (h(t_0) - h(t)) \bmod 24\big), \tag{S1.1}$$

where $d(t, t_0)$ is delay in hours between run time t and baseline time t_0, $h(\cdot)$ maps a time to hour-of-day, and mod24 indicates wrap-around on a 24-hour clock. The base comfort component uses the following piecewise schedule on delay hours d:

$$C_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - \frac{2}{3}\,d & 0 < d \leq 3 \\ 8 - \frac{1}{2}\,(d-3) & 3 < d \leq 7 \\ 6 - \frac{2}{5}\,(d-7) & 7 < d \leq 12 \\ 2 & d > 12 \end{cases} \tag{S1.2}$$

where $C_{\text{base}}(d)$ is the pre-penalty comfort score (0–1) as a function of delay d. The late-night noise penalty is applied when running between 10pm and 7am and the appliance exceeds a dBA threshold:

$$N(t, \text{type}, \text{housing}) = \begin{cases} 0 & \text{if } h(t) \in [7, 22) \text{ or } \text{dBA}(\text{type}) \leq \theta \\ 2 \cdot \kappa(\text{housing}) & \text{otherwise} \end{cases} \tag{S1.3}$$

where $N(t, \text{type}, \text{housing})$ is a nonnegative noise-disruption penalty in comfort-score points, dBA(type) is the appliance noise level for the appliance type, $\theta = 45\,\text{dBA}$ is the dBA threshold, and $\kappa(\text{housing})$ is a unitless housing-type multiplier (larger for shared-wall housing). Representative appliance levels are about 45 dBA for dishwashers (Bellingham Electric, 2026), 74 dBA for washers at spin-cycle peak (Precision Appliance Leasing, 2024), and 65 dBA for dryers (Coolblue B.V., 2026). The 45 dBA threshold sits above the EPA/WHO 35 dBA indoor night limit so that a dishwasher at its rated level is at threshold (U.S. Environmental Protection Agency, 1974), and only the $> 45\,\text{dBA}$ test reads the values, so washers and dryers incur the penalty regardless of the exact figure. The household-size penalty scales with delay relative to the appliance-specific maximum acceptable delay $d_{\max}(\text{type})$:

$$S(d, n, \text{type}) = \sigma(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})},\ 1\right), \tag{S1.4}$$

where $S(d, n, \text{type})$ is a nonnegative household-size penalty in comfort-score points, n is occupants, $d_{\max}(\text{type})$ is the maximum acceptable delay (hours) for the appliance type, and $\sigma(n)$ is an occupants-based factor (higher for larger households). Final comfort is then

$$C_{\text{comfort}} = \text{clip}_{[0,10]}\big(C_{\text{base}} - N - S\big) \tag{S1.5}$$

where $\text{clip}_{[a,b]}(x) = \min(b, \max(a, x))$ clamps a value to the interval $[a, b]$.

0.14.2 Shower Comfort

Comfort is a piecewise function peaking at the REU2016 average of 7.8 min (DeOreo, Mayer, Dziegielewski and Kiefer, 2016; Poll, 2024), with additive penalties for temperature adequacy (CDC Legionella thresholds (for Disease Control and Prevention, 2026), with 120 °F delivery and 140 °F storage following CDC guidance and 110 and 130 °F defined as comfort and scald boundaries, the latter near the 30-second deep-burn level of Moritz and Henriques (Moritz and Henriques, 1947)) and household contention. The implementation uses a duration-based piecewise base comfort $C_{\text{base}}(\tau)$ (with a short-duration cubic ramp to reflect severely rushed showers), then subtracts a temperature penalty $\Delta_{\text{temp}}(T_{\text{heater}})$ and a household contention penalty $\Delta_{\text{cont}}(\tau, n)$, and clips to $[0, 10]$. The comfort-optimal duration is the 7.8 min REU2016 mean at moderate outdoor temperatures but is shifted upward in cold weather to at most 10.3 min, from the 11.6 versus 8.8 min winter/summer field-mean ratio of Ibanez-Rueda et al. (2023), a southern-Spain sample, so the cap is a cross-population extrapolation (Ibáñez-Rueda, Guardiola, López-Ruiz and González-Gómez, 2023). This cold-weather elasticity is an explicit modeling assumption supported only in direction by the literature (Wong, Mui and Chan, 2022; DeOreo et al., 2016), and warm-weather scoring is unchanged:

$$C_{\text{comfort}}(\tau, T_{\text{heater}}, n) = \text{clip}_{[0,10]}\Big(C_{\text{base}}(\tau) - \Delta_{\text{temp}}(T_{\text{heater}}) - \Delta_{\text{cont}}(\tau, n)\Big). \tag{S1.6}$$

where τ is shower duration (min), n is occupants, $C_{\text{base}}(\tau)$ is the base comfort curve, $\Delta_{\text{temp}}(T_{\text{heater}})$ is a temperature-adequacy penalty, and $\Delta_{\text{cont}}(\tau, n)$ is a household-contention penalty.

0.15 Practicality

0.15.1 HVAC Practicality

Practicality starts at 10 and subtracts an asymmetric extremity penalty for setpoints beyond adoption-comfort bounds:

$$\delta(T_{\text{in}}) = \begin{cases} 1.0\,(71 - T_{\text{in}}) & \text{cooling, } T_{\text{in}} < 71 \\ 1.5\,(T_{\text{in}} - 82) & \text{cooling, } T_{\text{in}} > 82 \\ 1.8\,(63 - T_{\text{in}}) & \text{heating, } T_{\text{in}} < 63 \\ 0.8\,(T_{\text{in}} - 76) & \text{heating, } T_{\text{in}} > 76 \\ 0 & \text{otherwise} \end{cases} \tag{S1.7}$$

where $\delta(T_{\text{in}})$ is the extremity penalty (score points per °F beyond adoption-comfort bounds), with the steepest slopes on the least-tolerable directions. The pre-feasibility practicality score is $P_{\text{prac}} = \text{clip}_{[0.5,\,10]}\big(10 - \delta(T_{\text{in}})\big)$.

The result is then multiplied by an outdoor–indoor ΔT feasibility factor (1.0, 0.95, 0.85, 0.70 for $\Delta T < 10$, < 20, < 35, ≥ 35 °F), signaling operation near system limits. Finally, age- and maintenance-driven efficiency degradation enters here as a reliability factor: an older or poorly maintained unit is a less practical choice to rely on, so the score is scaled by $(1 - D)$ where

$$D = \min\Big(0.30,\ \delta_{\text{maint}}\big[\,1.5\min(a, 10) + 0.5\max(a - 10, 0)\,\big]\Big) \tag{S1.8}$$

is the fraction of efficiency lost (capped at 30%), a is HVAC age in years, and δ_{maint} is the base annual loss rate (0.005, 0.010, 0.015 per year for good, moderate, and poor upkeep); the first ten years degrade at $1.5\,\delta_{\text{maint}}$ and later years at $0.5\,\delta_{\text{maint}}$ (Domanski, Henderson and Payne, 2014). The final practicality score is clipped to $[0.5, 10]$.

0.15.2 Appliance Practicality

Practicality mirrors the delay schedule with additional penalties for late-night timing complexity and household coordination difficulty. The base practicality component is:

$$P_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - d & 0 < d \leq 2 \\ 8 - \frac{3}{4}(d-2) & 2 < d \leq 4 \\ 6.5 - \frac{1}{2}(d-4) & 4 < d \leq 8 \\ 4.5 - \frac{3}{8}(d-8) & 8 < d \leq 12 \\ 1.5 & d > 12 \end{cases} \tag{S1.9}$$

where $P_{\text{base}}(d)$ is the pre-penalty practicality score (0–1) as a function of delay d. Timing complexity applies an additive penalty by run-time hour, because low-price zones at the brink of the day are perceived as too early or too late to act on (Paetz et al., 2012):

$$T(t) = \begin{cases} 2 & 0 \leq h(t) < 6 \\ 1 & 22 \leq h(t) < 24 \\ 0 & \text{otherwise} \end{cases} \tag{S1.10}$$

where $T(t)$ is a nonnegative timing-complexity penalty in practicality-score points and $h(t)$ is hour-of-day. Household coordination difficulty uses the same delay scaling form as in comfort:

$$K(d, n, \text{type}) = \kappa_n(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})}, 1\right), \tag{S1.11}$$

where $K(d, n, \text{type})$ is a nonnegative household-coordination penalty in practicality-score points and $\kappa_n(n)$ is an occupants-based factor increasing with occupants. Final practicality is $C_{\text{prac}} = \text{clip}_{[1.5,10]}\big(P_{\text{base}} - T - K\big)$, with a daytime floor applied for $h(t) \in [7, 22)$ and $d \leq d_{\max}(\text{type})$.

0.15.3 Shower Practicality

Practicality additionally penalizes alternatives that exceed available tank capacity. Unlike comfort, the base practicality curve is not centered on the 7.8 min comfort optimum: it rises with duration up to a peak near 12 min before declining, because practicality here measures likelihood of adoption for everyone—how little behavior change an alternative demands of a household that already showers at a given length—rather than momentary preference. A very short shower scores low on practicality (it requires sustained restraint), while a moderately long one is easy to sustain. Pressure toward shorter showers in the MAVT aggregate therefore comes from the cost and environmental criteria and from the comfort optimum, not from practicality. Beyond the 15-minute Harris Poll threshold the tail declines conservatively: 33% of adults self-report showers longer than 15 min (Poll, 2024), while the metered REU2016 data implies a much lower true rate (DeOreo et al., 2016).

On the other hand, the capacity constraint below is what makes an over-long shower infeasible for a constrained household. The practicality model uses a duration-based base practicality $P_{\text{base}}(\tau)$ and a capacity constraint computed from hot-water demand. Hot-water per shower is $\tau\, \dot{V}\, f_{\text{hot}}$ (gal), total household hot-water need is $n\, \tau\, \dot{V}\, f_{\text{hot}}$, and available draw is approximated as 0.8 of nominal tank size S:

$$H_{\text{need}} = n\, \tau\, \dot{V}\, f_{\text{hot}}, \qquad H_{\text{avail}} = 0.8\, S. \tag{S1.12}$$

Table S2.1: Sensitivity analysis: Top-1 accuracy (%) by model and architecture under seven weight vectors.

	Gemini			DeepSeek			GPT-OSS			Qwen		
Weight vector	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$
Design (baseline)	36.2	47.2	93.1	36.7	54.5	90.8	33.3	48.0	91.6	30.0	47.0	89.7
Equal	50.0	55.3	94.9	36.8	52.2	93.7	37.7	49.4	93.5	28.8	47.2	89.5
Entropy pooled	37.6	48.8	92.7	35.1	53.2	89.1	33.8	48.0	90.6	29.7	46.1	87.6
Entropy per-type	38.5	52.1	94.6	34.3	55.0	92.4	34.6	48.9	92.9	30.4	45.4	93.1
MEREC pooled	**75.6**	74.2	92.2	43.5	58.3	93.0	46.9	53.5	90.9	32.3	49.9	94.5
MEREC per-type	69.1	76.6	94.4	49.5	52.9	94.1	41.4	57.1	92.8	34.5	51.2	92.4
MEREC HVAC vector	**80.4**	73.3	96.4	53.6	58.2	96.1	49.1	61.0	94.6	37.1	51.8	96.6

Percentage of the 195 test scenarios whose top-ranked alternative matches the reference. Bold marks cells in which $\mathcal{A}_{\mathrm{D}}$ exceeds $\mathcal{A}_{\mathrm{E}}$; these are the same two Gemini MEREC cells that reverse on Kendall's τ.

where H_{need} is total hot-water volume required for the household's showers over the relevant period, H_{avail} is available hot-water draw volume, and S is nominal tank capacity (all in gallons). A fixed penalty is applied when $H_{\mathrm{need}} > H_{\mathrm{avail}}$, yielding a clipped final practicality score.

S2. Extended Results and Robustness

This part of the supplementary material contains the full numerical and statistical results behind the main paper's Results section: per-model sensitivity tables, the complete RAG-ablation table, run-to-run variance distributions, statistical significance tests, and robustness checks of the failure-handling and aggregation protocols.

0.16 Full Sensitivity Analysis Results

Kendall's τ for every model, architecture and weight vector is reported in Section 4.7 of the main paper. Table S2.1 gives Top-1 accuracy for the same cells. The weight vectors are the design vector (0.30, 0.35, 0.20, 0.15), the equal-weight vector (0.25, 0.25, 0.25, 0.25), and the entropy- and MEREC-derived vectors of Appendix B, each applied pooled across the corpus and per decision type. No value in either table is pooled across models.

0.17 RAG Ablation: Full Per-Configuration Table

0.18 Run-to-Run Variance: Full Distributions

MAE and Top-1 distribution shapes across runs, and the per-model MAE breakdown, are omitted here because Table S2.10 (Appendix 0.27) already reports mean ± std for every metric.

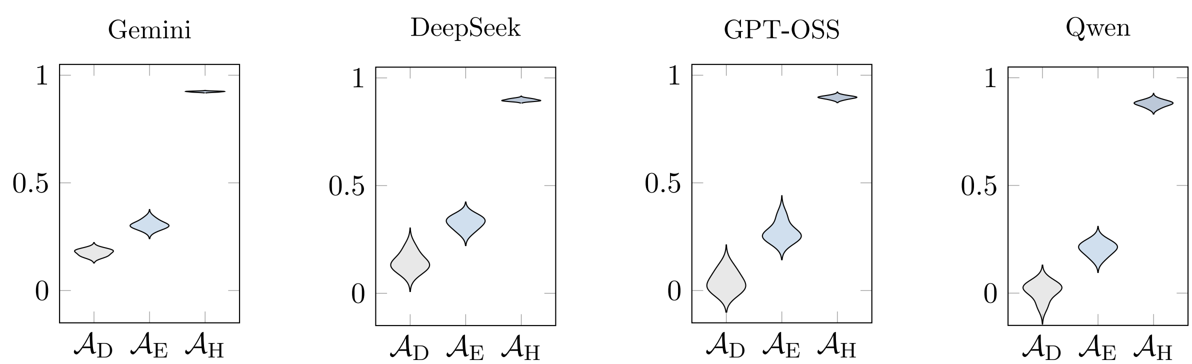
0.19 Statistical Significance Tests

Three complementary statistical approaches evaluate whether the architecture ordering ($\mathcal{A}_{\mathrm{H}} > \mathcal{A}_{\mathrm{E}} > \mathcal{A}_{\mathrm{D}}$) is statistically robust. All tests use per-scenario metrics computed from the 195 test scenarios, with five independent runs per architecture–model combination.

Configuration	Overall τ	95% CI	Model τ range	Overall MAE
Exemplars w/o hidden params	0.279	[0.200, 0.359]	[0.175, 0.391]	0.097
Control ($k = 3$, all-MiniLM-L6-v2)	0.277	[0.194, 0.356]	[0.154, 0.489]	0.098
Alt. embedding	0.246	[0.166, 0.324]	[0.165, 0.346]	0.101
Retrieval $k = 1$	0.242	[0.160, 0.325]	[0.182, 0.306]	0.104
Retrieval $k = 5$	0.230	[0.146, 0.311]	[0.112, 0.342]	0.091
Random exemplars ($k = 3$)	0.121	[0.044, 0.196]	[0.022, 0.276]	0.144
Descriptions w/o scores or ranks	0.068	[−0.011, 0.149]	[0.061, 0.073]	0.147
Nearest-neighbor (offline)	0.001	[−0.175, 0.172]	—	0.101

Evaluated on the full 90-scenario RAG corpus (35 HVAC / 35 Appliance / 20 Shower) under leave-one-out retrieval, with 3 models; Gemini is excluded on cost grounds. 95% CIs are percentile bootstrap intervals over scenario-level values. Model τ range spans the three per-model means. Nearest-neighbor assigns retrieved scenario scores directly without LLM rescoring and uses no LLM, so it has no per-model range. Across all LLM configurations 17,010 scoring calls were made with 6 failures (0.04%).

Table S2.2: RAG ablation: Kendall's τ and MAE by configuration.



0.20 Intraclass Correlation Coefficient

Table S2.3 reports ICC(2,1) values decomposing total metric variance into between-model and residual components.

0.21 Pairwise Wilcoxon Tests with Stouffer Combination

Wilcoxon signed-rank tests compare each pair of adjacent architectures ($\mathcal{A}_D$ vs. $\mathcal{A}_E$ and $\mathcal{A}_E$ vs. $\mathcal{A}_H$) on each of the three primary metrics (τ, MAE, Top-1), applied independently within each of the four models and combined across models using Stouffer's method. All 24 per-model comparisons on these three metrics and all 6 Stouffer-combined comparisons reach $p_{\text{Holm}} < .001$ within the 56-test family described in Section 3.9 of the main paper. That family's one non-significant test—DeepSeek $\mathcal{A}_D$ vs. $\mathcal{A}_E$ on the RMSE/MAE ratio, $p_{\text{Holm}} = 0.093$—is on a metric not shown here.

0.22 Effect Sizes

Table S2.4 reports Cliff's δ for each pairwise comparison.

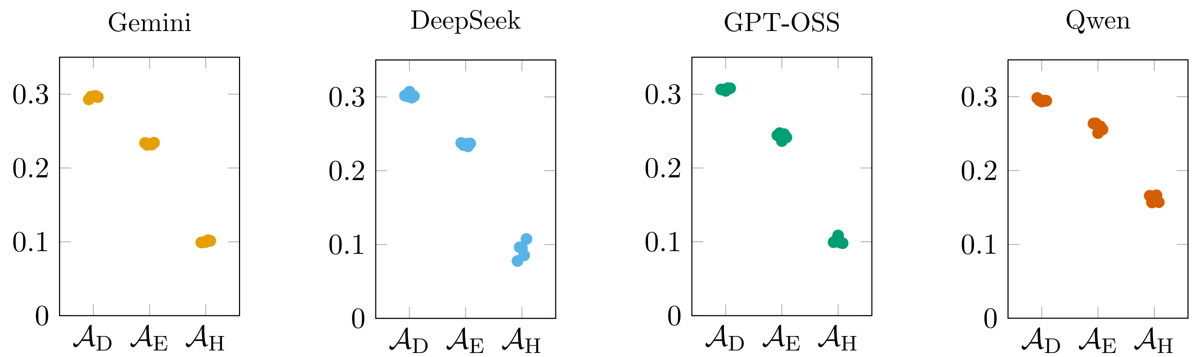


Figure S2.2: Strip plots of Overall RMSE across five runs.

Table S2.3: ICC(2,1) variance decomposition: fraction of total variance attributable to between-model differences (σ^2_{model}) and residual ($\sigma^2_{\text{residual}}$).

Architecture	Metric	σ^2_{model}	$\sigma^2_{\text{residual}}$
$\mathcal{A}_\mathrm{D}$	Kendall's τ	0.427	0.573
	MAE	0.814	0.186
	Top-1	0.081	0.919
$\mathcal{A}_\mathrm{E}$	Kendall's τ	0.276	0.724
	MAE	0.829	0.171
	Top-1	0.171	0.829
$\mathcal{A}_\mathrm{H}$	Kendall's τ	0.448	0.552
	MAE	0.807	0.193
	Top-1	0.267	0.733

0.23 Bootstrap Confidence Intervals

0.24 Prompt Ablation Significance

Table S2.6 reports the Friedman omnibus tests for $\mathcal{A}_\mathrm{E}$ referenced in Section 4.9 of the main paper, and Table S2.7 reports the pairwise comparisons that reach significance under the two-stage Holm correction (Section 3.7 of the main paper) with a fully reported effect size.

0.25 Imputed Robustness Check

The main results (Section 4 of the main paper) use per-run-then-average aggregation: failed scenarios are excluded from their run's metric computation, and metrics are averaged across five runs. To test sensitivity to this exclusion, Table S2.8 reports an imputed alternative at per-run granularity: within each run, scenarios that carry a failed score receive the scale midpoint (0.5) on the affected criterion scores before that run's metrics are computed, and every run evaluates the full 195-scenario test set.

The $\mathcal{A}_\mathrm{H} > \mathcal{A}_\mathrm{E} > \mathcal{A}_\mathrm{D}$ ordering survives per-run imputation: pooled τ equals 0.871, 0.279, and 0.094 for the three architectures, and each of the four models preserves the ordering. The largest movement occurs on GPT-OSS $\mathcal{A}_\mathrm{H}$, where Kendall's τ falls from 0.897 to 0.786 (-0.111),

Table S2.4: Cliff's δ effect sizes for pairwise architecture comparisons, shown per model.

Comparison	Model	τ	MAE	Top-1
$\mathcal{A}_{\mathrm{D}}$ vs. $\mathcal{A}_{\mathrm{E}}$	DeepSeek	−0.182 (small)	0.504 (large)	−0.270 (small)
	Gemini	−0.128 (negligible)	0.402 (medium)	−0.143 (negligible)
	GPT-OSS	−0.275 (small)	0.487 (large)	−0.217 (small)
	Qwen	−0.244 (small)	0.338 (medium)	−0.317 (small)
$\mathcal{A}_{\mathrm{E}}$ vs. $\mathcal{A}_{\mathrm{H}}$	DeepSeek	−0.737 (large)	0.867 (large)	−0.589 (large)
	Gemini	−0.592 (large)	0.766 (large)	−0.510 (large)
	GPT-OSS	−0.788 (large)	0.832 (large)	−0.666 (large)
	Qwen	−0.829 (large)	0.737 (large)	−0.725 (large)

Table S2.5: Bootstrap 95% BCa confidence intervals for primary metrics per model (10,000 resamples).

Metric	Model	$\mathcal{A}_{\mathrm{D}}$ (95% CI)	$\mathcal{A}_{\mathrm{E}}$ (95% CI)	$\mathcal{A}_{\mathrm{H}}$ (95% CI)
Kendall's τ	DeepSeek	0.144 [0.119, 0.182]	0.328 [0.301, 0.348]	**0.897 [0.894, 0.902]**
	Gemini	0.176 [0.165, 0.186]	0.305 [0.294, 0.321]	**0.923 [0.921, 0.925]**
	GPT-OSS	0.041 [0.013, 0.083]	0.270 [0.248, 0.325]	**0.897 [0.891, 0.901]**
	Qwen	0.010 [-0.029, 0.032]	0.207 [0.182, 0.227]	**0.880 [0.868, 0.888]**
MAE	DeepSeek	0.232 [0.230, 0.234]	0.167 [0.165, 0.168]	**0.047 [0.043, 0.050]**
	Gemini	0.220 [0.218, 0.221]	0.159 [0.158, 0.160]	**0.048 [0.048, 0.049]**
	GPT-OSS	0.242 [0.240, 0.243]	0.169 [0.166, 0.171]	**0.052 [0.052, 0.054]**
	Qwen	0.235 [0.234, 0.236]	0.191 [0.188, 0.195]	**0.072 [0.071, 0.074]**
Top-1 (%)	DeepSeek	36.7 [35.0, 40.4]	54.5 [52.3, 55.8]	**90.8 [90.3, 91.9]**
	Gemini	36.2 [35.4, 37.6]	47.2 [46.2, 49.9]	**93.1 [92.9, 93.3]**
	GPT-OSS	33.3 [31.7, 35.2]	48.0 [46.1, 50.6]	**91.6 [91.2, 92.5]**
	Qwen	30.0 [27.4, 32.0]	47.0 [45.0, 49.0]	**89.7 [89.1, 91.2]**

Top-1 falls from 91.6% to 83.6%, and MAE rises from 0.052 to 0.078. GPT-OSS imputes 23.4 scenarios per run on $\mathcal{A}_{\mathrm{H}}$ (12% of its test set), which drives the movement. Imputation lowers $\mathcal{A}_{\mathrm{H}}$'s success-conditioned metrics, so failure exclusion inflated them; pooled $\mathcal{A}_{\mathrm{H}}$ τ falls from 0.899 to 0.871 (−0.029). Every other cell moves by at most 0.006.

0.26 Aggregation Method Sensitivity

Table S2.9 compares the per-run-then-average protocol (Method A) used throughout the main text with a mean-aggregate-then-evaluate protocol (Method C) in which per-scenario scores are averaged across all five runs before metric computation. Method C produces a consensus ranking that may differ from the average of per-run rankings.

$\mathcal{A}_{\mathrm{H}}$ differences are negligible ($|\Delta\tau| \leq 0.010$ across all models). $\mathcal{A}_{\mathrm{E}}$ differences are small for three models but reach 0.104 for GPT-OSS. For $\mathcal{A}_{\mathrm{D}}$, Method C produces slightly higher τ (up to +0.03) because averaging scores across disagreeing runs can synthesize a ranking that no individual run produces. The architecture ordering is preserved in all cases.

Table S2.6: Prompt ablation: Friedman omnibus tests for $\mathcal{A}_{\mathrm{E}}$, corrected across the full 12-test family (both architectures, all three models, both metrics).

Model	Metric	χ^2_2	raw p	p_{Holm}
DeepSeek	Kendall's τ	1.51	0.470	0.922
GPT-OSS	Kendall's τ	3.38	0.185	0.922
Qwen	Kendall's τ	5.71	0.058	0.460
DeepSeek	Top-1	35.71	1.8×10^{-8}	2.1×10^{-7}
Qwen	Top-1	9.87	0.007	0.065

None of the Kendall's τ omnibus tests survive correction; the Top-1 omnibus is significant only for DeepSeek. Holm adjustment is step-down with monotonicity enforced, so a test with a larger raw p cannot receive a smaller adjusted p: DeepSeek and GPT-OSS Kendall's τ therefore share the adjusted value 0.922. The GPT-OSS Top-1 omnibus is part of the 12-test family but is not reported here.

Table S2.7: Prompt ablation: significant pairwise comparisons (Holm-corrected post-hoc, run only within omnibus families that survived correction).

Arch.	Model	Comparison	Value	p_{Holm}	Cliff's δ
$\mathcal{A}_{\mathrm{E}}$	DeepSeek	control vs. cot_scaffold (Top-1)	$0.552 \to 0.367$	2.9×10^{-7}	0.276 (small)
$\mathcal{A}_{\mathrm{E}}$	DeepSeek	control vs. scale_0_10 (Top-1)	$0.552 \to 0.478$	0.003	0.116 (negligible)
$\mathcal{A}_{\mathrm{E}}$	DeepSeek	cot_scaffold vs. scale_0_10 (Top-1)	$0.367 \to 0.478$	2.2×10^{-4}	−0.190 (small)
$\mathcal{A}_{\mathrm{D}}$	Qwen	control vs. cot_scaffold (τ)	$-0.052 \to 0.092$	3.8×10^{-4}	−0.165 (small)
$\mathcal{A}_{\mathrm{D}}$	Qwen	control vs. scale_0_10 (τ)	$-0.052 \to 0.015$	0.028	−0.100 (negligible)

All five comparisons are significant under the two-stage Holm correction (Section 3.7 of the main paper). Cliff's δ sign follows $P(\mathrm{config}_i > \mathrm{config}_j) - P(\mathrm{config}_i < \mathrm{config}_j)$; negative values indicate the second-listed configuration scores higher.

0.27 Overall Results: Numerical Values

Table S2.10 reports the full numerical results for each metric, architecture, and model, with five-run means and standard deviations. Table S2.11 provides a MAE breakdown by criterion, showing each architecture's best- and worst-performing model.

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Table S2.8: Comparison of per-run-then-average (Method A) vs per-run imputed (0.5 midpoint) aggregation. Values are means across all four models.

Architecture	Metric	Method A	Imputed (per-run)
$\mathcal{A}_{\text{D}}$	τ	0.093	0.094
$\mathcal{A}_{\text{D}}$	MAE	0.232	0.232
$\mathcal{A}_{\text{D}}$	Top-1 (%)	34.1	34.2
$\mathcal{A}_{\text{E}}$	τ	0.277	0.279
$\mathcal{A}_{\text{E}}$	MAE	0.171	0.172
$\mathcal{A}_{\text{E}}$	Top-1 (%)	49.2	49.3
$\mathcal{A}_{\text{H}}$	τ	0.899	0.871
$\mathcal{A}_{\text{H}}$	MAE	0.055	0.061
$\mathcal{A}_{\text{H}}$	Top-1 (%)	91.3	89.2

Mean scenarios imputed per run: DeepSeek $\mathcal{A}_{\text{D}}$ 1.4, DeepSeek $\mathcal{A}_{\text{H}}$ 0.2, Gemini $\mathcal{A}_{\text{E}}$ 4.0, GPT-OSS $\mathcal{A}_{\text{H}}$ 23.4, Qwen $\mathcal{A}_{\text{D}}$ 0.4, Qwen $\mathcal{A}_{\text{H}}$ 0.4; all other cells impute none. GPT-OSS imputes 20–26 scenarios per run on $\mathcal{A}_{\text{H}}$, 12% of its test set.

Imputed scenarios receive 0.5 on all four criterion scores; their ranks follow the pipeline's deterministic tie-break (appearance order).

Table S2.9: Method A (per-run) vs Method C (mean-aggregate) for each architecture and model.

Architecture	Model	Method A τ	Method C τ	$\lvert\Delta\tau\rvert$
$\mathcal{A}_{\text{D}}$	Gemini	0.176	0.169	0.007
$\mathcal{A}_{\text{D}}$	DeepSeek	0.144	0.159	0.015
$\mathcal{A}_{\text{D}}$	GPT-OSS	0.041	0.053	0.012
$\mathcal{A}_{\text{D}}$	Qwen	0.010	−0.002	0.012
$\mathcal{A}_{\text{E}}$	Gemini	0.305	0.296	0.009
$\mathcal{A}_{\text{E}}$	DeepSeek	0.328	0.350	0.023
$\mathcal{A}_{\text{E}}$	GPT-OSS	0.270	0.374	0.104
$\mathcal{A}_{\text{E}}$	Qwen	0.207	0.244	0.038
$\mathcal{A}_{\text{H}}$	Gemini	0.923	0.925	0.001
$\mathcal{A}_{\text{H}}$	DeepSeek	0.897	0.897	0.000
$\mathcal{A}_{\text{H}}$	GPT-OSS	0.897	0.887	0.010
$\mathcal{A}_{\text{H}}$	Qwen	0.880	0.879	0.001

	Gemini			DeepSeek		
Metric	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$
Kendall's τ	0.176 ± 0.014	0.305 ± 0.019	**0.923 ± 0.002**	0.144 ± 0.041	0.328 ± 0.029	**0.897 ± 0.005**
Top-1 (%)	36.2 ± 1.5	47.2 ± 2.1	**93.1 ± 0.3**	36.7 ± 3.1	54.5 ± 2.3	**90.8 ± 0.9**
Top-2 (%)	69.7	79.4	**98.5**	72.2	81.5	**97.7**
MAE	0.220 ± 0.002	0.159 ± 0.001	**0.048 ± 0.001**	0.232 ± 0.002	0.167 ± 0.002	**0.047 ± 0.004**
RMSE	0.296 ± 0.002	0.233 ± 0.001	**0.100 ± 0.001**	0.302 ± 0.003	0.236 ± 0.002	**0.092 ± 0.011**
RMSE/MAE	1.35 ± 0.00	1.47 ± 0.01	2.09 ± 0.01	1.30 ± 0.01	1.41 ± 0.01	1.97 ± 0.07

	GPT-OSS			Qwen		
Metric	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$
Kendall's τ	0.041 ± 0.045	0.270 ± 0.042	**0.897 ± 0.007**	0.010 ± 0.039	0.207 ± 0.030	**0.880 ± 0.013**
Top-1 (%)	33.3 ± 2.4	48.0 ± 2.8	**91.6 ± 0.8**	30.0 ± 3.0	47.0 ± 2.7	**89.7 ± 1.2**
Top-2 (%)	66.5	76.9	**98.1**	67.7	76.8	**96.9**
MAE	0.242 ± 0.002	0.169 ± 0.003	**0.052 ± 0.001**	0.235 ± 0.002	0.191 ± 0.004	**0.072 ± 0.002**
RMSE	0.307 ± 0.002	0.243 ± 0.004	**0.101 ± 0.004**	0.295 ± 0.002	0.259 ± 0.006	**0.162 ± 0.005**
RMSE/MAE	1.27 ± 0.01	1.44 ± 0.01	1.93 ± 0.04	1.26 ± 0.01	1.35 ± 0.00	2.24 ± 0.02

Table S2.10: Overall performance by model (5-run mean ± std, 195 scenarios; Top-2 accuracy is a single run-averaged value, not tracked with std). Bolded values indicate the best architecture per metric per model.

	$\mathcal{A}_{\mathrm{D}}$		$\mathcal{A}_{\mathrm{E}}$		$\mathcal{A}_{\mathrm{H}}$	
Criterion	Gemini	GPT-OSS	Gemini	Qwen	DeepSeek	Qwen
Energy Cost	0.248	0.258	0.178	0.195	0.068	0.112
Environmental	0.248	0.292	0.167	0.192	0.081	0.150
Comfort	0.171	0.185	0.138	0.193	0.017	0.011
Practicality	0.212	0.231	0.152	0.186	0.021	0.017
Overall MAE	0.220	0.242	0.159	0.191	0.047	0.072

For each architecture, the left column is the best-performing model and the right column is the worst, identified by 5-run mean Overall MAE. Environmental MAE for Shower reflects water-consumption proxy (gallons); see Section 3.4 of the main paper.

Table S2.11: MAE per criterion by architecture and model.

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